LESSON 10 · CHEAT SHEET

pandas II: Pivots & Lookups

COUNTING

```
df["dept"].value_counts()            # COUNTIF, sorted
df["dept"].value_counts(normalize=True)  # shares
df["dept"].nunique()                 # distinct values
pd.crosstab(df["quartile"], df["gender"])
pd.crosstab(a, b, normalize="index") # % per row
```

GROUPBY = PIVOTTABLE

```
g = df.groupby("dept")["salary"]  # rows, values
g.mean()   g.median()   g.sum()   g.count()
g.agg(["count", "mean", "median", "max"])
g.sum().sort_values(ascending=False)
g.mean().idxmax()                  # which row is max

df.groupby(["dept", "gender"])["hourly"].mean()
   .unstack()        # 2nd key -> columns
df.groupby("dept").agg(
    people=("employee_id", "count"),
    avg=("salary", "mean"))
```

PIVOT_TABLE

```
df.pivot_table(index="dept",      # Rows box
               columns="gender",  # Columns box
               values="hourly",   # Values box
               aggfunc="mean",    # Summarise by
               margins=True,      # Grand Total
               fill_value=0)      # empty cells
aggfunc=["mean", "count"]         # two summaries
```

MERGE = XLOOKUP

```
df.merge(other, on="dept", how="left")
# left:  keep all of df, NaN if no match
# inner: keep matches only (rows vanish!)
# outer: keep everything from both
df.merge(other, left_on="dept", right_on="name")
len(df)   # check before AND after
```

MISSING VALUES (NAN)

```
df.isna().sum()              # per column
df["cc"].isna().sum()        # one column
df[df["cc"].isna()]          # the rows
df["cc"] = df["cc"].fillna("CC-999")
df = df.dropna(subset=["cc"])
df["bonus"].mean()           # ignores NaN
```

QUARTILES & BINS

```
pd.qcut(df["hourly"], 4, labels=[...])  # equal count
pd.cut(df["age"], [0, 30, 50, 99])      # fixed edges
df.groupby("quartile", observed=True)["hourly"]
  .agg(["min", "max"])
```

PICKING CELLS FROM A TABLE

```
t = g.agg(["mean", "median"])
t.loc["M", "mean"]           # one cell
t["mean"]                    # one column
t.loc["HR"]                  # one row
t.reset_index()              # index -> column
```

EXCEL → PANDAS

COUNTIF	`value_counts()`
PivotTable	`groupby("a")["v"].mean()`
rows + columns	`pivot_table(...)`
Grand Total	`margins=True`
XLOOKUP	`merge(how="left")`
#N/A	`NaN`
IFERROR(x, 0)	`.fillna(0)`
QUARTILE groups	`pd.qcut(col, 4)`

GOTCHAS

- Two keys: `groupby(["a", "b"])`, a list
- `how="inner"` drops unmatched rows silently; count rows
- A groupby result is a Series: `.unstack()` or `.reset_index()` to get a table
- Round at the end, not before grouping
- NaN is not 0: `.fillna(0)` only when 0 is true